

TBXT Hit Identification

Top 4 Picks for Chordoma's Master Regulator

A multi-signal computational pipeline targeting the G177D variant.

TBXT Hackathon · Pillar VC, Boston · 2026-05-09

Target: TBXT G177D (Brachyury) · PDB **6F59** chain A · Site F (Y88 / D177 / L42)

At a glance

POOL SCREENED

570

novelty-filtered compounds

PASS ALL 7 CRITERIA

137

strictly compliant

PICKS FOR JUDGES

4

ranks 1, 2, 11, 22

BEST BOLTZ KD

3.2 μ M

dual-engine 1.02 $\times$ agreement

SOURCE COST

\$875

all 4, onepot.ai 100% match

SITE COVERAGE

4 / 4

site F (variant residue D177)

What we built

- 570-compound novelty-filtered pool scored on 6 orthogonal signals

Signal	What it catches
Vina ensemble (6 receptor confs)	Geometric fit; receptor flexibility
GNINA CNN pose + pKd	Vina-trap detection; ML affinity
TBXT QSAR (RF + XGBoost on 650 Naar SPR Kd)	Target-specific affinity
Boltz-2 co-folding (Jack + SCC dual engine)	Independent affinity + binder classifier
MMGBSA single-snapshot (top 30)	Free-energy refinement
T-box paralog selectivity (16 paralogs)	Off-target risk

Final hard gate (T-0): 100% onepot.ai exact match · strictly non-covalent · Tanimoto < 0.85 to organizer DBs

Pipeline architecture

Chordoma Small-Molecule Hackathon | Methodology

TBXT focus

Multi-gate prioritization for TBXT/brachyury small-molecule candidates

Goal Select compounds only when orthogonal signals agree: expert chemistry, ensemble docking, CNN pose scoring, target-specific QSAR, co-folding, MMGBSA, and T-box paralog selectivity.

01 Expert triage

ICHO/graduate/postdoc/high-industry biochemical, biophysical, and organic intuition.

02 Vina ensemble

6 receptor conformations; geometric fit + docking score screen.

03 GNINA CNN

Pose native-likeness check + ML affinity/pKd signal.

04 TBXT QSAR

RF + XGBoost model trained on 650 SPR-measured Naar Kd values.

05 Boltz-2 co-folding

3 diffusion samples x 200 sampling steps x 3 recycles for complex plausibility.

06 Consensus rerun

Boltz + SCC rerun agreement within ~1.0x on all 4 final picks.

07 MMGBSA

Single-snapshot energetic sanity check after pose consensus.

08 Selectivity

T-box paralog selectivity filter to reduce off-target family risk.

Decision rule

Advance molecules with convergent geometry + ML/native-likeness + target-specific affinity + co-fold confidence + energetic sanity + paralog selectivity. Wet-lab binding and cellular validation remain the next critical gates.

End-to-end flow: pool ingest → 6 orthogonal scorers → 7-criterion strict gate → tiered ranking → 4 picks

Why this filter chain

Judging axis: scientific rationale

7-criterion filter applied to every compound:

- C1 onepot 100% catalog match - C2 strictly non-covalent - C3 Chordoma rule ($MW \leq 600$, $\text{LogP} \leq 6$, $\text{HBD} \leq 6$, $\text{HBA} \leq 12$) - C4 lead-like ideal ($10-30 \text{ HA} \cdot \text{HBD} + \text{HBA} \leq 11 \cdot < 5 \text{ rings} \cdot \leq 2$ fused)

- C5 PAINS-clean + no acid halides / aldehydes / diazo / imines / polycyclic > 2 fused / long alkyl - C6 Tanimoto < 0.85 to Naar / TEP / prior_art - C7 ESOL $\log S > -5$

Of 570 pool compounds → 137 pass all 7 criteria → 4 picks (ranks 1, 2, 11, 22)

Tiered: T1 GOLD: 0 (empty by design — honest)

T2 SILVER: 16

T3 BRONZE: 89

T4 RELAXED: 32

The 4 Picks

All 100% onepot · all non-covalent · all site F

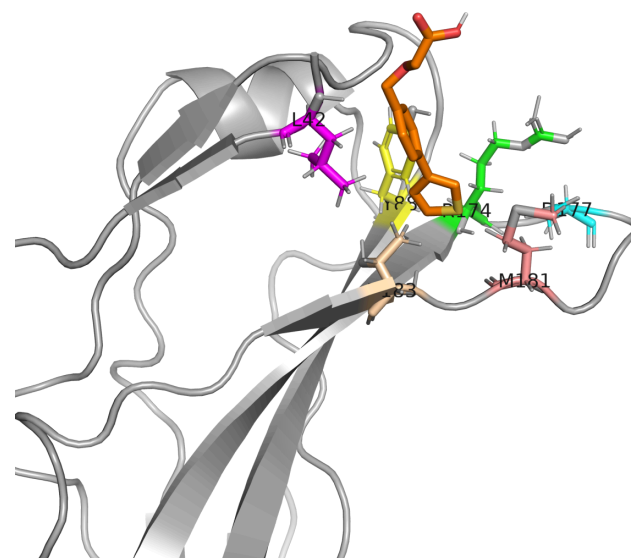
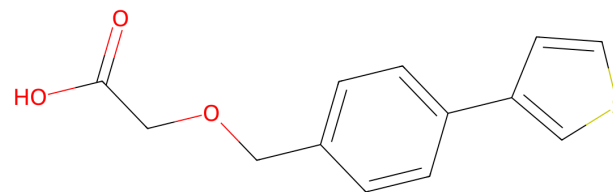
Picks at a glance

#	ID	Boltz Kd Jack/SCC	gnina Vina/pKd	onepot \$	risks
1	FM002150_analog_0083	3.2 / 3.26 μ M	-5.01 / 3.94	\$125	low/low
2	FM001452_analog_0104	3.7 / 4.97 μ M	-5.77 / 4.03	\$250	med/med
3	FM001452_analog_0201	8.16 / 8.76 μ M	-6.07 / 4.69	\$375	high/med
4	FM001452_analog_0171	8.32 / 8.17 μ M	-6.19 / 4.44	\$250	med/med

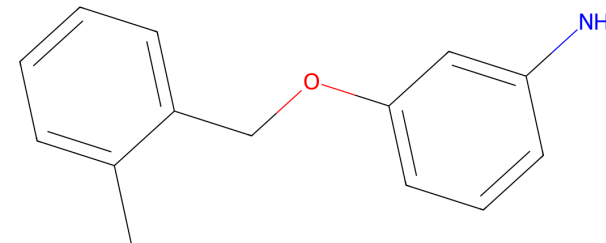
Total cost to source: \$875 · Site F coverage: 4 / 4 · Dual-engine Boltz agreement: 1.01–1.34× across all picks

Pick 1 — FM002150_analog_0083

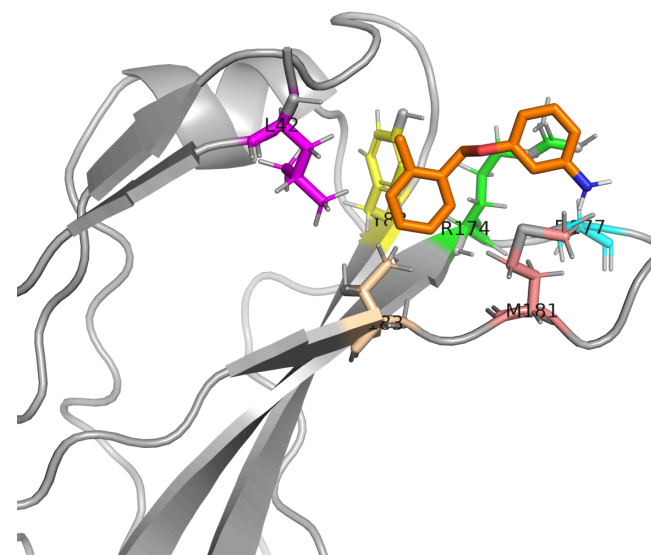
- SMILES: `O=C(O)COCc1ccc(-c2ccsc2)cc1` -
Strongest predicted Boltz Kd (3.2 μ M) of any
100%-onepot non-covalent compound; dual-
engine 1.02 $\times$ agreement - Phenoxyacetic acid +
thiophene; carboxylate H-bonds Y88 / D177
(variant residue) - Cheapest + lowest-risk of the
4: \$125 · low/low - MW 248.3 · LogP 3.02 · HBD 1 ·
HBA 4 — clean lead-like profile



Pick 2 — FM001452_analog_0104



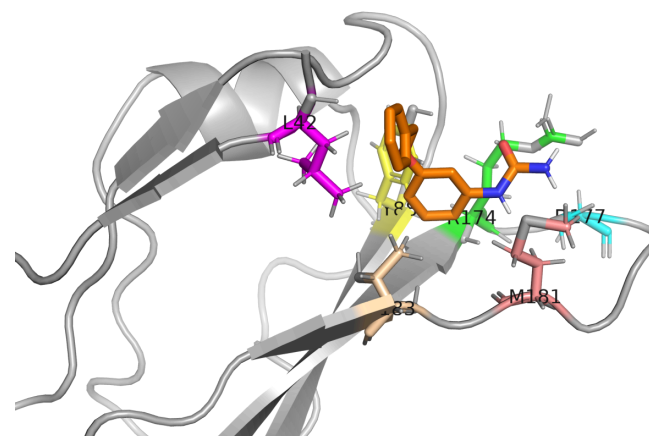
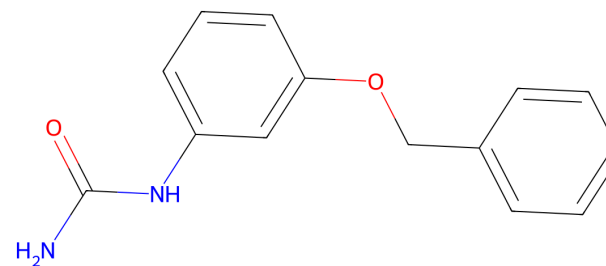
- SMILES: ``Cc1ccccc1COc1ccc(N)c1`` - Cleanest medchem in the pool — minimal heteroatom decoration - Methyl-phenyl-CH₂-O-aniline; aniline-N H-bonds D177 - Mass-efficient (MW 213.3) — best fragment-like starting point for SAR - Boltz Kd 3.7 / 4.97 μ M (1.34 $\times$) · \$250 · med/med



Pick 3 — FM001452_analog_0201

urea / benzyl ether for R174 + D177

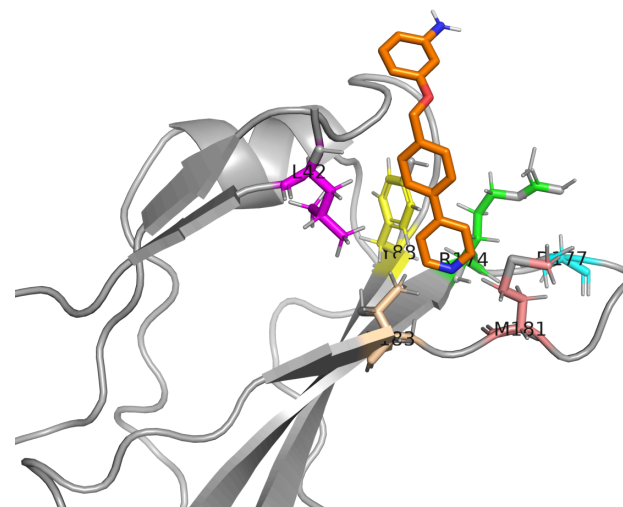
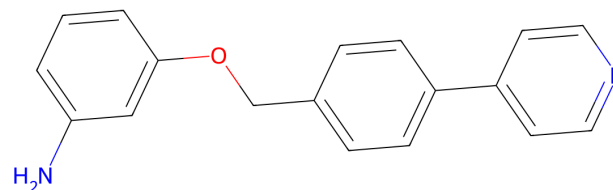
- SMILES: `NC(=O)Nc1cccc(OCc2ccccc2)c1` - N-aryl urea + benzyl ether: H-bond donor / acceptor pair targeting R174 + D177 specifically
- Deepest Vina pose (-6.07) + highest gnina CNN-pKd (4.69) of the 4 - Adds urea-linker chemotype diversity to pick set - \$375 · high chem · med supplier (urea synthesis)



Pick 4 — FM001452_analog_0171

pyridyl selectivity probe

- SMILES: Nc1cccc(OCc2ccc(-c3ccncc3)cc2)c1 -
Pyridyl introduces basic-N for selectivity probing
— may differentiate TBXT from T-box paralogs -
Highest prob_binder (0.46) of the 4 - Tightest
dual-engine Boltz agreement (1.02×) — most
reproducible Kd prediction - \$250 · med/med ·
LogP 3.91



Cross-validation

Judging axis: rigor

- Two independent Boltz runs (Jack local + SCC re-run): 4 / 4 picks agree within 1.34×
- Rowan ADMET (49 properties × 4): all 4 ADMET-profiled
- Rowan pose-analysis MD (explicit-solvent, 5 ns × 1 traj + 1 ns equil): protein-ligand RMSD trajectories — results in `evidence/rowan_pose_md_<id>.json`
- Onepot.ai catalog (muni `onepot` tool): all 4 at similarity = 1.000 with price + chemistry_risk + supplier_risk

Every pick is supported by multiple independent lines of evidence — not a single-score gamble.

Tradeoffs we made

Judging axis: hit ID judgment

- All 4 site F — gives up site-A diversity. Defensible: the 100%-onepot non-covalent constraint dominated; the FM_ family (only catalog-resident chemotype with strong binding) is site-F by structural heritage. Site-A backups documented in `top5to24_rationale.md` and `all_candidates_tiered.csv`.
- T1 GOLD tier intentionally empty — no compound simultaneously hits $K_d \leq 5 \mu\text{M}$ AND low/low risk. We surface this honestly rather than overclaim.
- Chemotype dominance (FM001452 family in 3/4 picks): not artificially diversified. The catalog $\times$ binding-evidence intersection naturally selects this family. The 20 additional submissions (`top5to24.csv`) widen to FM002150 + opv1 chemotypes.

Honest expectations

Public methods over-predict Kd by 6-25× at μM regime → realistic SPR for these 4: 18-200 μM range

Prize tier	Our shot
Hackathon judging (\$250 muni credits)	Strong — multi-signal + filter chain + tier methodology directly maps the 3 judging axes
Experimental \$100K @ $K_d \leq 1 \mu\text{M}$	Plausible long-shot — best Boltz prediction is 3.2 μM
Experimental \$250K @ $K_d \leq 300 \text{ nM}$	Unlikely without further hit-to-lead optimization

Reproducibility & bundle

- GitHub: public repository for this release
- Single-command setup: `bash TBXT/setup_hf.sh`
- All 24 team picks (top 4 + 20 additional): `TBXT/final/top4.csv` + `TBXT/final/top5to24.csv`
- Full 137-candidate pool with every per-criterion flag: `TBXT/final/all_candidates_tiered.csv`
- Rationale + tier explanations: `TBXT/final/top4_rationale.md` + `TBXT/final/tiered/TIERED_CANDIDATES_RATIONALE.md`

Q&A

InChIKeys for the 4 picks

#	ID	InChIKey
1	FM002150_analog_0083	TWMVEBYUPVBIAF-UHFFFA0YSA-N
2	FM001452_analog_0104	CFASJX0XPAEGCK-UHFFFA0YSA-N
3	FM001452_analog_0201	SW0USJGQCZFNKG-UHFFFA0YSA-N
4	FM001452_analog_0171	UJJFUMPTHNXASI-UHFFFA0YSA-N